

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
VI Semester: TEST-1 (Section: F)

20IS610- Statistical Methods in Information Processing

Duration :1 Hr.

21/03/2024

Max. Marks:20

Note:

- I. Both Part A and Part B are Compulsory
- II. Attend any one from Part A and Part B

Q.NO	CO	CD	QUESTION	MARKS
PART A				
1.a	CO2	Apply	Marks secured by 50 students in test paper are given below: 30 45 48 55 39 25 31 12 18 21 54 59 51 33 43 44 10 38 19 26 41 35 37 41 46 33 51 37 58 58 17 19 23 26 29 38 57 36 35 44 43 27 19 43 22 31 47 34 31 15 Prepare a frequency table with class intervals 10-19, 20-29, 30-39, ... and calculate the value of Arithmetic Mean from the following table obtained.	8M
1.b	CO1	Remembering	List any two mathematical properties of Arithmetic Mean	2M
OR				
2.a	CO2	Apply	Calculate Mean and Standard Deviation from the following data Value: 90-99 80-89 70-79 60-69 50-59 40-49 30-39 Freq : 2 12 22 20 14 4 1	8M
2.b	CO2	Remembering	Explain the various measures of central tendency	2M
PART B				
3.a	CO2	Apply	Find the Mean Deviation from Median for the following data: Class interval: 0-10 10-20 20-30 30-40 40-50 50-60 60-70 Freq : 8 12 10 8 3 2 7	8M
3.b	CO1	Remembering	List out the various measures of dispersion	2M
OR				

4.a	CO2	Apply	Find the average rate of increase in the population which in the first decade had increased by 20%, in the next by 30% and in the third by 40%.	
4.b	CO2	Apply	Find the Geometric Mean for the following: Marks: 0-10 10-20 20-30 30-40 40-50 No. : 5 7 15 25 8 students	6M

Course Outcome: At the end of the course the students will have the ability to	
CO-1	Explain the different statistical measures and analysis types available
CO-2	Apply a selected method to solve an unknown problem
CO-3	Analyze results and make inferences

--- End ---

JSN

20IS610

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
 VI Semester: TEST-1 (Section: F)

20IS610- Statistical Methods in Information Processing

Duration : 1 Hr.

17/04/2023

Max. Marks:20

Note:

- I. Both Part A and Part B are Compulsory
 II. Attend any one from Part A and Part B

Q.NO	CO	CD	QUESTION	MARKS
PART A				
1.a	CO2	Apply	Marks secured by 50 students in test paper are given below: 30 45 48 55 39 25 31 12 18 21 54 59 51 33 43 44 10 38 19 26 41 35 37 41 46 33 51 37 58 58 17 19 23 26 29 38 57 36 35 44 43 27 19 43 22 31 47 34 31 15 Prepare a frequency table with class intervals 10-19, 20-29, 30-39, ... and calculate value of Arithmetic Mean from following table obtained	8M
1.b	CO1	Remember	List any two mathematical properties of Arithmetic Mean	2M
OR				
2.a	CO2	Apply	Calculate Mean and Standard Deviation from the following data Value: 90-99 80-89 70-79 60-69 50-59 40-49 30-39 Freq : 2 12 22 20 14 4 1	8M
2.b	CO2	Apply	Calculate Pearson's Co-efficient of Skewness: Mean= 48.6, Median=39, S.D= 39.5	2M
PART B				
3.a	CO2	Apply	Find the Mean Deviation from Median for the following data: Class interval: 0-10 10-20 20-30 30-40 40-50 50-60 60-70 Freq : 8 12 10 8 3 2 7	8M

3.b	CO1	Apply	Consider the following distribution			JSS SCIENCE	
			Distribution A				Distribution B
			Mean	100			90
			Median	90			80
			SD	10			10
I) Is the distribution A has same degree of variation as distribution B II) State True/ False whether both distribution have same degree of skewness							
CR							
4.a	CO2	Apply	Find the range and Co-efficient of range for the following observations 65, 70, 82, 59, 81, 76, 57, 60, 55, 50			2M	
4.b	CO2	Apply	Calculate first 4 moments about mean ^{median} for following data X: 1 2 3 4 5 6 7 8 9 F: 1 6 13 25 30 22 9 5 2			8M	

Course Outcome: At the end of the course the students will have the ability to

CO-1	Explain the different statistical measures and analysis types available
CO-2	Apply a selected method to solve an unknown problem
CO-3	Analyze results and make inferences

--- End ---

20IS610

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
VI Semester: TEST-2 (Section: F)

20IS610- Statistical Methods in Information Processing

Duration :1 Hr.

02/05/2024

Max. Marks:20

Note:

- I. Part A is Compulsory.
- II. Attend any one from Part B.

Q.NO	CO	CD	QUESTION	MARKS																																																
PART A																																																				
1	CO2	Apply	Calculate the first four moments about the mean for the following data and comment on the nature of the distribution: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>f</td><td>1</td><td>6</td><td>13</td><td>25</td><td>30</td><td>22</td><td>9</td><td>5</td><td>2</td></tr></table>	x	1	2	3	4	5	6	7	8	9	f	1	6	13	25	30	22	9	5	2	10M																												
x	1	2	3	4	5	6	7	8	9																																											
f	1	6	13	25	30	22	9	5	2																																											
PART B																																																				
2.a	CO2	Apply	Find the Rank Correlation Co-efficient from the following marks awarded by the examiners in statistics <table><tr><th>Roll No.</th><th>Examiner A</th><th>Examiner B</th><th>Examiner C</th></tr><tr><td>1</td><td>24</td><td>37</td><td>30</td></tr><tr><td>2</td><td>29</td><td>35</td><td>28</td></tr><tr><td>3</td><td>19</td><td>16</td><td>20</td></tr><tr><td>4</td><td>14</td><td>26</td><td>25</td></tr><tr><td>5</td><td>30</td><td>23</td><td>25</td></tr><tr><td>6</td><td>19</td><td>27</td><td>30</td></tr><tr><td>7</td><td>27</td><td>19</td><td>20</td></tr><tr><td>8</td><td>30</td><td>20</td><td>24</td></tr><tr><td>9</td><td>20</td><td>16</td><td>22</td></tr><tr><td>10</td><td>28</td><td>11</td><td>29</td></tr><tr><td>11</td><td>11</td><td>21</td><td>15</td></tr></table>	Roll No.	Examiner A	Examiner B	Examiner C	1	24	37	30	2	29	35	28	3	19	16	20	4	14	26	25	5	30	23	25	6	19	27	30	7	27	19	20	8	30	20	24	9	20	16	22	10	28	11	29	11	11	21	15	8M
Roll No.	Examiner A	Examiner B	Examiner C																																																	
1	24	37	30																																																	
2	29	35	28																																																	
3	19	16	20																																																	
4	14	26	25																																																	
5	30	23	25																																																	
6	19	27	30																																																	
7	27	19	20																																																	
8	30	20	24																																																	
9	20	16	22																																																	
10	28	11	29																																																	
11	11	21	15																																																	
2.b	CO2	Understand	Explain the difference between Karl Pearson's correlation co-efficient and rank correlation co-efficient	2M																																																
OR																																																				
3.a	CO2	Apply	From the following data, obtain the two regression equations:	6M																																																

			Sales:	91	97	108	121	67	124	51	73	111	57
			Purchases:	71	75	69	97	70	91	39	61	80	47
3.b	CO1	Understand	Explain the methods of Measurement of Trend										

Course Outcome: At the end of the course the students will have the ability to

CO-1	Explain the different statistical measures and analysis types available												
CO-2	Apply a selected method to solve an unknown problem												
CO-3	Analyze results and make inferences												

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
 VI Semester: TEST-3 (Section: F)

20IS610- Statistical Methods in Information Processing

Duration :1 Hr.

19/06/2023

Max. Marks 20

Q.NO	CO	CD	QUESTION	MARKS																														
PART A																																		
1.	CO3	Apply	To test the hypothesis that the average number of days a patient is kept in the three local hospitals say A,B and C is same, a random check on the number of days that seven patients stayed in each hospital reveals the following Hospital A: 8 5 9 2 7 8 2 Hospital B: 4 3 8 7 7 1 5 Hospital C: 1 4 9 8 7 2 3	10M																														
PART B																																		
2.	CO3	Apply	Using "Ratio to Trend" method, determine the quarterly seasonal indices for the following data <table border="1"> <thead> <tr> <th>Year</th><th>I Qtr</th><th>II Qtr</th><th>III Qtr</th><th>IV Qtr</th></tr> </thead> <tbody> <tr> <td>1</td><td>68</td><td>60</td><td>61</td><td>63</td></tr> <tr> <td>2</td><td>70</td><td>58</td><td>56</td><td>60</td></tr> <tr> <td>3</td><td>68</td><td>63</td><td>68</td><td>67</td></tr> <tr> <td>4</td><td>65</td><td>56</td><td>56</td><td>62</td></tr> <tr> <td>5</td><td>60</td><td>55</td><td>55</td><td>58</td></tr> </tbody> </table>	Year	I Qtr	II Qtr	III Qtr	IV Qtr	1	68	60	61	63	2	70	58	56	60	3	68	63	68	67	4	65	56	56	62	5	60	55	55	58	10M
Year	I Qtr	II Qtr	III Qtr	IV Qtr																														
1	68	60	61	63																														
2	70	58	56	60																														
3	68	63	68	67																														
4	65	56	56	62																														
5	60	55	55	58																														
OR																																		
3.	CO3	Apply	Test the hypothesis that the median HDL cholesterol levels in adult population of city A and B are same. Use the following observations and use Mann-Whitney test at 0.05 level of significance (5) City A: 42 20 51 39 57 60 23 City B: 30 42 25 29 35	10M																														

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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
VI Semester: TEST-3 (Section: F)

20IS610- Statistical Methods in Information Processing

Duration :1 Hr.

06/06/2024

Max. Marks:20

Q.NO	CO	CD	QUESTION	MARKS																														
PART A																																		
1.	CO3	Apply	A company's trainees are randomly divided into 3 groups of 10 each and are given a course in management skills by three different methods. At the end of the training period, they are given a test and their scores are as follows: Method A : 99 64 101 85 79 88 97 95 90 100 Method B: 83 102 125 61 91 96 94 89 93 75 Method C: 89 98 56 105 87 90 87 101 76 89 Use the kruskal-wallis test to determine at 5% level of significance if the three methods are equally effective[Given critical value is 5.991]	10M																														
PART B																																		
2.	CO3	Apply	Using "Ratio to Trend" method, determine the quarterly seasonal indices for the following data <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Year</th><th>I Qrt</th><th>II Qrt</th><th>III Qrt</th><th>IV Qrt</th></tr> </thead> <tbody> <tr> <td>1</td><td>68</td><td>60</td><td>61</td><td>63</td></tr> <tr> <td>2</td><td>70</td><td>58</td><td>56</td><td>60</td></tr> <tr> <td>3</td><td>68</td><td>63</td><td>68</td><td>67</td></tr> <tr> <td>4</td><td>65</td><td>56</td><td>56</td><td>62</td></tr> <tr> <td>5</td><td>60</td><td>55</td><td>55</td><td>58</td></tr> </tbody> </table>	Year	I Qrt	II Qrt	III Qrt	IV Qrt	1	68	60	61	63	2	70	58	56	60	3	68	63	68	67	4	65	56	56	62	5	60	55	55	58	10M
Year	I Qrt	II Qrt	III Qrt	IV Qrt																														
1	68	60	61	63																														
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3	68	63	68	67																														
4	65	56	56	62																														
5	60	55	55	58																														
OR																																		
3.	CO3	Apply	A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have an average thickness of 0.024 cm with standard deviation of 0.002 cm. Test the significance of the deviation. Value of t at the 5% level is 2.262.	6M																														
4.	CO3	Apply	List and Explain different seasonal variations methods	4M																														

JSS MAHAVIDYAPEETHA
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VI Semester BE Degree Examination
 Department of Information Science and Engineering

STATISTICAL METHODS IN INFORMATION PROCESSING

Duration: 3 Hours

Max. Marks:100

NOTE: PART A is compulsory.
PART B has internal choice.

PART – A

Q.No.	CO	CD	PI	QUESTION	MARKS												
1.	CO2	Apply	1.2.1	1 a) In a frequency distribution of 100 families given below, the number of families corresponding to expenditure groups 20-40 and 60-90 are missing from the table. However, the median is known to be 50. Find the missing frequencies. <table border="1"> <tr> <td>Expenditure</td><td>0-20</td><td>20-40</td><td>40-60</td><td>60-80</td><td>80-100</td></tr> <tr> <td>No. of families</td><td>14</td><td>?</td><td>27</td><td>?</td><td>15</td></tr> </table>	Expenditure	0-20	20-40	40-60	60-80	80-100	No. of families	14	?	27	?	15	06
Expenditure	0-20	20-40	40-60	60-80	80-100												
No. of families	14	?	27	?	15												
				1 b) Given that mean of a distribution is 5, variance is 9 and the moment co-efficient of Skewness is -1, find the first three moments about the origin.	04												
2.	CO1	Apply	1.2.1	2 a) For 100 students of a class, the regression equation of marks in stats(X) on the marks in commerce (Y) is $3Y - 5X + 180 = 0$. The mean marks in commerce are 50 and the variance of marks in stats is $\frac{4}{9}$th of the variance of the marks in commerce. Find the mean marks in statistics and the co-efficient of correlation between the marks in 2 subjects.	06												
				2 b) Differentiate between Correlation Analysis and Regression Analysis.	04												

3.	CO2	Apply	1.2.1	The decision-making body of fertilizer firm producing fertilizers wants to predict future sales trend for the year 2006 and 2008 based on the analysis of its past sales pattern. The sales of the firm for last 7 year for this purpose are given below: <table border="1"> <tr> <td>Year</td> <td>1998</td> <td>1999</td> <td>2000</td> <td>2001</td> <td>2002</td> <td>2003</td> <td>2004</td> </tr> <tr> <td>Sales</td> <td>70</td> <td>75</td> <td>90</td> <td>98</td> <td>85</td> <td>91</td> <td>100</td> </tr> </table>	Year	1998	1999	2000	2001	2002	2003	2004	Sales	70	75	90	98	85	91	100	10
Year	1998	1999	2000	2001	2002	2003	2004														
Sales	70	75	90	98	85	91	100														
4.	CO3	Apply	1.2.1	a) Godrej soap manufacturing company was distributing a particular brand of soap through a large number of retail shops. Before a heavy advertisement campaign, the mean sales per week per shop were 140 dozens. After the campaign, a sample of 26 shops was taken and the mean sales were found to be 147 dozens with standard deviation 16. Can the advertisement be effective? Given that $t_{25}(0.05)=1.708$.	08																
				b) Define ANOVA and Explain the types of ANOVA.	02																
5.	CO1	Understand	1.2.1	Explain the different tools used for Statistical Quality Control.	10																

PART - B

Q.No.	CO	CD	PI	QUESTION	MARKS																
6	CO1	Apply	1.2.1	a) From the following data of income distribution, calculate the Arithmetic Mean. It is given that: I) Total income of persons in the highest group is Rs.435. II) None is earning less than Rs.20. <table border="1"> <thead> <tr> <th>Income (Rs)</th> <th>No. of persons</th> </tr> </thead> <tbody> <tr><td>Below 30</td><td>16</td></tr> <tr><td>Below 40</td><td>36</td></tr> <tr><td>Below 50</td><td>61</td></tr> <tr><td>Below 60</td><td>76</td></tr> <tr><td>Below 70</td><td>87</td></tr> <tr><td>Below 80</td><td>95</td></tr> <tr><td>80 and over</td><td>5</td></tr> </tbody> </table> b) Mention and explain the Mathematical properties of Arithmetic Mean.	Income (Rs)	No. of persons	Below 30	16	Below 40	36	Below 50	61	Below 60	76	Below 70	87	Below 80	95	80 and over	5	06
Income (Rs)	No. of persons																				
Below 30	16																				
Below 40	36																				
Below 50	61																				
Below 60	76																				
Below 70	87																				
Below 80	95																				
80 and over	5																				
OR					04																

7.	CO1	Apply	1.2.1	7.a) From the following data, calculate Moments about assumed mean 25 and convert them into Central Moments (First four moments). <table><tr><td>CI</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td></tr><tr><td>F</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table>	CI	0-10	10-20	20-30	30-40	F	1	3	4	2	06						
				CI	0-10	10-20	20-30	30-40													
F	1	3	4	2																	
				7.b) Calculate Karl Pearson's co-efficient of Skewness from the following data: <table><tr><td>Size</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Freq</td><td>10</td><td>18</td><td>30</td><td>25</td><td>12</td><td>3</td><td>2</td></tr></table>	Size	1	2	3	4	5	6	7	Freq	10	18	30	25	12	3	2	04
Size	1	2	3	4	5	6	7														
Freq	10	18	30	25	12	3	2														

8.	CO2	Apply	1.2.1	8.a) From the following table calculate Co-efficient of Correlation by Karl Pearson's method. <table><tr><td>X</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>Y</td><td>9</td><td>11</td><td>A</td><td>8</td><td>7</td></tr></table> Find the value of A, given that Arithmetic Mean of X and Y series are 6 and 8 respectively.	X	6	2	10	4	8	Y	9	11	A	8	7	08
				X	6	2	10	4	8								
Y	9	11	A	8	7												
				8.b) List and explain any two properties of Correlation Co-efficient.	02												

OR

✓	CO2	Apply	1.2.1	Ten Competitors in beauty contest are ranked by 3 judges in the following order:	10																																				
				<table border="1"><tr><td>1st judge</td><td>A</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>2nd judge</td><td>B</td><td>3</td><td>5</td><td>8</td><td>4</td><td>7</td><td>10</td><td>2</td><td>1</td><td>6</td><td>9</td></tr><tr><td>3rd judge</td><td>C</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table>		1 st judge	A	1	6	5	10	3	2	4	9	7	8	2 nd judge	B	3	5	8	4	7	10	2	1	6	9	3 rd judge	C	6	4	9	8	1	2	3	10	5	7
				1 st judge		A	1	6	5	10	3	2	4	9	7	8																									
				2 nd judge		B	3	5	8	4	7	10	2	1	6	9																									
3 rd judge	C	6	4	9	8	1	2	3	10	5	7																														
Use Rank Correlation Co-efficient to determine which pair of Judges has the nearest to common tastes in beauty.																																									

10.	CO3	Apply	4.6.3	Fit a Straight line Trend to the following data:								10
				Year	1965	1966	1967	1968	1969	1970	1971	

				Gross ex- factory value	672	824	967	1204	1464	1758	2057
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					OR	
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Compute the seasonal index for the following data assuming that there is no need to adjust data for the trend.

Qtr	2000	2001	2002	2003	2004
I	100	110	120	130	140
II	110	120	130	140	150
III	120	130	140	150	160
IV	130	140	150	160	170

11.	CO3	Apply	4.6.3	Qrt	2000	2001*	2002	2003	2004	2005
				I	3.5	3.5	3.5	4.0	4.1	4.2
				II	3.9	4.1	3.9	4.6	4.4	4.6
				III	3.4	3.7	3.7	3.8	4.2	4.3
				IV	3.6	4.8	4.0	4.5	4.5	4.7

III	34	37	39	46	44	46	10
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A farmer applied 3 types of fertilizers on 4 separate plots. The figures on yield per acre are tabulated below:

Fertilizer	Yield
1	10
2	15
3	20

Fertilizer s plots	Yield			
	A	B	C	D
Nitrogen	6	4	8	6
Potash	7	6	6	9
Phosphat es	8	5	10	9

Find out if plots are materially different in fertility and if the 3 fertilizers make any material difference in yields. Given that $F(2,6)$ and $F(3,6)$ at 5% alpha are 5.14 and 4.76 respectively.

[illegible]

					OR	
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13	CO3	Apply	1.2.1	13.a) Test the hypothesis that median HDL cholesterol levels in the adult population of city A and city B are the same. Use the following observations and use Mann Whitney test at 0.05 level of significance. Given that critical value is 5.
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critical value is 5

13.b) Explain the procedure of Testing of Hypothesis.

[illegible]

Q4.	CO2	Apply	4.6.3	The following figures give the number of defectives in 20 samples, each sample containing 2000 items.
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10

425, 430, 216, 341, 225, 322, 280, 306, 337, 305, 356, 402, 216, 264, 126, 409, 193, 326, 280, 389.

Calculate the values for the central line and the control limits for p-chart. Draw the p-chart and comment if the process can be regarded in control or not?

OR

A machine is set to deliver packets of a given weight. 10 samples of size 5 each were recorded. Below are given relevant data:

Sample No	1	2	3	4	5	6	7	8	9	10
Mean	15	17	15	18	17	14	18	15	17	16
Range	7	7	4	9	8	7	12	4	11	5

Calculate the values for the central line and control limits of mean chart and the range and then comment on the state of control.

(Conversion factors for $n=5$ are $A_2 = 0.58$, $D_3 = 0$, $D_4 = 2.115$)

10

Course Outcome (CO):

CO-1	Explain the different statistical measures and analysis types available
CO-2	Apply a selected method to solve an unknown problem
CO-3	Analyze results and make inferences

Performance Indicator (PI):

1.2.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems.
4.6.3	Represent data (in tabular and/or graphical forms) so as to facilitate analysis and explanation of the data, and drawing of conclusions.

--- End ---

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

VI Semester BE Degree Examination
Department of Information Science and Engineering
Statistical Methods in Information Processing

Duration: 3 Hours

Max. Marks:100

NOTE: Answer all questions in Part-A
Part-B has internal choices

PART – A

Q.NO	CO	CD	PI	QUESTION	MARKS																						
1.	CO1	Apply	1.1.1	✓ 1 a) Calculate the Mean Deviation from Median for the following data <table><tr><td>Marks less than</td><td>80</td><td>70</td><td>60</td><td>50</td><td>40</td><td>30</td><td>20</td><td>10</td></tr><tr><td>No. students</td><td>100</td><td>90</td><td>80</td><td>60</td><td>32</td><td>20</td><td>13</td><td>5</td></tr></table>	Marks less than	80	70	60	50	40	30	20	10	No. students	100	90	80	60	32	20	13	5	6				
				Marks less than	80	70	60	50	40	30	20	10															
No. students	100	90	80	60	32	20	13	5																			
				✓ 1 b) The mean of marks in statistics of 100 students in a class is 72. The mean of marks of boys is 75, while their number was 70. Find out the mean marks of girls in the class	4																						
2.	CO2	Apply	1.1.1	From the data given find a) The two Regression co-efficient b) The 2-regression equation c) Co-efficient of correlation between the marks in Economics and Statistics d) Most likely marks in Statistics when marks in Economics is 30 <table><tr><td>Marks in Eco</td><td>25</td><td>28</td><td>35</td><td>32</td><td>31</td><td>36</td><td>29</td><td>38</td><td>34</td><td>32</td></tr><tr><td>Marks in Stats</td><td>43</td><td>46</td><td>49</td><td>41</td><td>36</td><td>32</td><td>31</td><td>30</td><td>33</td><td>39</td></tr></table>	Marks in Eco	25	28	35	32	31	36	29	38	34	32	Marks in Stats	43	46	49	41	36	32	31	30	33	39	10
Marks in Eco	25	28	35	32	31	36	29	38	34	32																	
Marks in Stats	43	46	49	41	36	32	31	30	33	39																	

✓ 3.	CO2	Apply	1.1.1	The decision-making body of fertilizers firm producing fertilizers wants to predict future sales trend for the year 2006 and 2008 based on the analysis of its past sales pattern. The sales of the firm for last 7 year for this purpose are given below: <table><tr><td>Year</td><td>1998</td><td>1999</td><td>2000</td><td>2001</td><td>2002</td><td>2003</td><td>2004</td></tr><tr><td>Sales</td><td>70</td><td>75</td><td>90</td><td>98</td><td>85</td><td>91</td><td>100</td></tr></table>	Year	1998	1999	2000	2001	2002	2003	2004	Sales	70	75	90	98	85	91	100	10																																																	
Year	1998	1999	2000	2001	2002	2003	2004																																																															
Sales	70	75	90	98	85	91	100																																																															
4.	CO2	Apply	1.1.1	A manufacturer wishes to determine the effectiveness of four types of machines (A, B, C and D) in the production of bolts. To accumulate this, the numbers of defective bolts produced for each of two shifts in the results are shown in the following table: <table><tr><th rowspan="2">Machine</th><th colspan="5">First Shift</th><th colspan="5">Second Shift</th></tr><tr><th>M</th><th>T</th><th>W</th><th>TH</th><th>F</th><th>M</th><th>T</th><th>W</th><th>TH</th><th>F</th></tr><tr><td>A</td><td>6</td><td>4</td><td>5</td><td>5</td><td>4</td><td>5</td><td>7</td><td>4</td><td>6</td><td>8</td></tr><tr><td>B</td><td>10</td><td>8</td><td>7</td><td>7</td><td>9</td><td>7</td><td>9</td><td>12</td><td>8</td><td>8</td></tr><tr><td>C</td><td>7</td><td>5</td><td>6</td><td>5</td><td>9</td><td>9</td><td>7</td><td>5</td><td>4</td><td>6</td></tr><tr><td>D</td><td>8</td><td>4</td><td>6</td><td>5</td><td>5</td><td>5</td><td>7</td><td>9</td><td>7</td><td>10</td></tr></table> Perform an analysis of variance to determine at 5% level of significance, whether there is a difference (a) Between the machines and (b) Between the shifts. Given that F at 3 and 32 degrees of freedom is 2.90	Machine	First Shift					Second Shift					M	T	W	TH	F	M	T	W	TH	F	A	6	4	5	5	4	5	7	4	6	8	B	10	8	7	7	9	7	9	12	8	8	C	7	5	6	5	9	9	7	5	4	6	D	8	4	6	5	5	5	7	9	7	10	10
Machine	First Shift					Second Shift																																																																
	M	T	W	TH	F	M	T	W	TH	F																																																												
A	6	4	5	5	4	5	7	4	6	8																																																												
B	10	8	7	7	9	7	9	12	8	8																																																												
C	7	5	6	5	9	9	7	5	4	6																																																												
D	8	4	6	5	5	5	7	9	7	10																																																												
5.	CO3	Apply	1.1.1	The following data gives the number of defectives in 10 independent samples of varying samples from a production process <table><tr><th>Sample no</th><th>Sample size</th><th>No. of defectives</th></tr><tr><td>1</td><td>2000</td><td>425</td></tr><tr><td>2</td><td>1500</td><td>430</td></tr><tr><td>3</td><td>1400</td><td>216</td></tr><tr><td>4</td><td>1350</td><td>341</td></tr><tr><td>5</td><td>1250</td><td>225</td></tr><tr><td>6</td><td>1760</td><td>322</td></tr><tr><td>7</td><td>1875</td><td>280</td></tr></table>	Sample no	Sample size	No. of defectives	1	2000	425	2	1500	430	3	1400	216	4	1350	341	5	1250	225	6	1760	322	7	1875	280	10																																									
Sample no	Sample size	No. of defectives																																																																				
1	2000	425																																																																				
2	1500	430																																																																				
3	1400	216																																																																				
4	1350	341																																																																				
5	1250	225																																																																				
6	1760	322																																																																				
7	1875	280																																																																				

20

8	1955	306
9	3125	337
10	1575	305

Draw the control chart for fraction defective and comment on it.

10

8	CO1	Apply	1.1.1	8.a) From the following table calculate Co-efficient of Correlation by Karl Pearson's method <table border="1"> <tr> <td>X</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr> <tr> <td>Y</td><td>9</td><td>11</td><td>A</td><td>8</td><td>7</td></tr> </table> Find the value of A, given that Arithmetic Mean of X and Y series are 6 and 8 respectively.	X	6	2	10	4	8	Y	9	11	A	8	7	8
X	6	2	10	4	8												
Y	9	11	A	8	7												
				8.b) List and explain any two properties of Correlation Co-efficient.	2												

OR

9	CO1	Apply	1.1.1	Ten Competitors in beauty contest are ranked by 3 judges in the following order <table><tr><td>1st judge</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>2nd judge</td><td>3</td><td>5</td><td>8</td><td>4</td><td>7</td><td>10</td><td>2</td><td>1</td><td>6</td><td>9</td></tr><tr><td>3rd judge</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table> Use rank correlation Co-efficient to determine which pair of Judges has the nearest to common tastes in beauty	1 st judge	1	6	5	10	3	2	4	9	7	8	2 nd judge	3	5	8	4	7	10	2	1	6	9	3 rd judge	6	4	9	8	1	2	3	10	5	7	10
1 st judge	1	6	5	10	3	2	4	9	7	8																												
2 nd judge	3	5	8	4	7	10	2	1	6	9																												
3 rd judge	6	4	9	8	1	2	3	10	5	7																												

10	CO	Apply	1.1.1	Compute the seasonal indices for the following time series of sales (in thousands) of a commodity by the ratio to trend method:					10
				Year	I Quarter	II Quarter	III Quarter	IV Quarter	
				2008	800	920	880	820	
				2009	540	760	680	620	
				2010	400	580	540	480	
				2011	340	520	500	440	
				2012	300	400	360	340	

OR

11	CO	Apply	1.1.1	Compute the seasonal index for the following data assuming that there is no need to adjust data for the trend	10
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14	CO3	Apply	1.1.1	A machine is set to deliver packets of a given weight 10 samples of size 5 each were recorded. Below are given relevant data: <table border="1"> <tr> <td>Sample no</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> <td>8</td> <td>9</td> <td>10</td> </tr> <tr> <td>Mean</td> <td>15</td> <td>17</td> <td>15</td> <td>18</td> <td>17</td> <td>14</td> <td>18</td> <td>15</td> <td>17</td> <td>16</td> </tr> <tr> <td>Range</td> <td>7</td> <td>7</td> <td>4</td> <td>9</td> <td>8</td> <td>7</td> <td>12</td> <td>4</td> <td>11</td> <td>5</td> </tr> </table> Calculate the values for the central line and the control limits of mean chart and the range and then comment on the state of control. Given that $A_2=0.58, D_3=0, D_4=2.115$	Sample no	1	2	3	4	5	6	7	8	9	10	Mean	15	17	15	18	17	14	18	15	17	16	Range	7	7	4	9	8	7	12	4	11	5	10
				Sample no	1	2	3	4	5	6	7	8	9	10																								
				Mean	15	17	15	18	17	14	18	15	17	16																								
				Range	7	7	4	9	8	7	12	4	11	5																								

OR

The following figures give the number of defectives in 20 samples, each sample containing 2000 items

425, 430, 216, 341, 225, 322, 280, 306, 337, 305,
356, 402, 216, 264, 126, 409, 193, 326, 280, 389

Calculate the values for central line and control limits for p-chart. Draw the p-chart and comment if the process can be regarded in control or not

USN															
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22IS610

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

VI Semester BE Degree Examination
 Department of Information Science and Engineering

STATISTICAL METHODS IN INFORMATION PROCESSING

Duration: 3 Hours

Max. Marks:100

NOTE: Answer all the questions. Part A is compulsory Part B has internal choices.

PART - A

PART - A																								
Q.NO	CO	CD	PI	MARKS																				
	CO 1	Apply	1.1.1	10																				
	1. Explain the concept of Skewness and Kurtosis and also Calculate the 1 st 4 moments about mean for the following data																							
	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>f</td><td>1</td><td>6</td><td>13</td><td>25</td><td>30</td><td>22</td><td>9</td><td>5</td><td>2</td></tr></table>				x	1	2	3	4	5	6	7	8	9	f	1	6	13	25	30	22	9	5	2
x	1	2	3	4	5	6	7	8	9															
f	1	6	13	25	30	22	9	5	2															
	CO 2	Apply	1.1.1	10																				
	2. A data analyst is studying the interdependence between three measurable factors; labeled X1, X2 and X3 represent scores obtained from three different diagnostic tools used to evaluate student performance in a training program. The analyst wants to understand how well two of these variables can predict the third. You are task is with helping the analyst perform a multiple correlation analysis																							
	i)Calculate the Pearson correlation coefficients between each pair of variables																							
	ii)Using the correlation coefficients, Calculate the multiple correlation coefficient																							
	<table><tr><td>X1</td><td>2</td><td>5</td><td>7</td><td>11</td></tr><tr><td>X2</td><td>3</td><td>6</td><td>10</td><td>12</td></tr><tr><td>X3</td><td>1</td><td>3</td><td>6</td><td>10</td></tr></table>	X1	2	5	7	11	X2	3	6	10	12	X3	1	3	6	10								
X1	2	5	7	11																				
X2	3	6	10	12																				
X3	1	3	6	10																				
	CO 3	Apply	1.1.1	10																				
	3. A district sales office of a leading fast moving consumer goods company has been tracking the yearly sales of one of its flagship products. The company's management team is looking to make informed strategic resource allocation. To support this, they want to understand the long term trend in sales performance. The data team has been asked to analyze the historical sales figures																							
	and Solve by using least square method																							

	Year	2000	2001	2002	2003	2004	2005
Sugar prod	40	45	46	42	47	50	46

CO 4	Apply	1.1.1	10
4. A Manufacturer claims that a special type of projector bulb has as average life of 160 hrs. To check this, claim an investigator takes a sample of 20 such bulbs, puts on the test and obtains average life 167 hrs with SD 16 hrs. Assuming that life time of such bulbs follows Normal Distribution, the critical values were 2.093. Does the investigator accept the manufacturers claim at 5% level of significance?			

CO 5		Apply		1.1.1		10					
A machine is set to deliver packets of given weight. 10 samples of size 5 each were recorded. Below are relevant data. Calculate values for central line and control limits of mean chart and range. $A_2=0.58$, $D_3=0$, $D_4=2.115$											
5.	Sam ple no.	1	2	3	4	5	6	7	8	9	10
	$\bar{x}$	15	17	15	18	17	14	18	15	17	16
	R	7	7	4	9	8	7	12	4	11	5

PART - B

Q.NO	CO	CD	PI	MARKS			
6.	CO 1	Apply	1.1.1	10			
	Calculate mean deviation from median distribution.						
	CI	50-100	100-150	150-200	200-250	250-300	300-350
	F	7	18	25	31	15	4

OR

7.	CO 1	Apply	1.1.1	10
	Define arithmetic mean and explain the mathematical properties of arithmetic mean. By using same technique solve: The mean of marks in statistics of 100 students in a class was 72. The mean of marks of boys was 75, while their number was 70. Find out the mean marks of girls in the class			

[illegible]

CO 2	Apply	4.2.2	10
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Discuss the importance of Regression. The heights of fathers and sons in inches are given. Find 2 lines of regression. One for predicting the height of the son based on fathers' height and other for predicting height of the father based on son's height. Estimate the average height of a son when height of the father is 68.5 inch.

Height of father	65	66	67	67	68	69	70	71
Height of son	66	68	65	69	74	73	72	70

OR

CO 2	Apply	4.2.2	10
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Differentiate between correlation and regression. Calculate the rank correlation coefficient from the following data

X	78	89	97	69	59	79	68
Y	125	137	156	112	107	136	124

CO 3	Apply	1.1.1	10
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Compute the seasonal indices by the ratio to trend method for the data of quarterly demand of electricity

Year	summer	monsoon	winter	spring
2019	70	52	22	31
2020	101	64	24	45
2021	120	75	30	49
2022	135	82	34	50

OR

CO 3	Apply	4.2.2	10
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11. Explain the various methods of measuring trends. Solve by applying semi-average method where a mid-sized manufacturing company had recorded its annual sales over 6-year period as follows. The company management wants to analyze past performance to improve future sales strategies as a part of analysis team

	Year	1982	1983	1984	1985	1986	
	Sales	10	12	11	11	15	25

CO 4

Apply

4.2.2

10

A farmer applied 3 types of fertilizers on 4 separate plots. The figures on the yield per acre are tabulated below. Given that $F(2,6)=5.14$ and $F(3,6)=4.76$ at 5% level of significance. Find out if plots are materially differ in fertility, as also, if the 3 fertilizers make any material differ in yields.

Fertilizers plots	Yield			
	A	B	C	D
Nitrogen	6	4	8	6
Potash	7	6	6	9
Phosphate	8	5	10	9

OR

CO 4

Apply

1.1.1

10

In a study, 12 participants were divided into 3 groups of 4 each, they were subjected to 3 different conditions, A (Low noise), B (Average noise), C (Loud noise). They were given a test and errors committed by them on test were noted and are given in table below. Given that critical value is 5.692

Participant No.	Condition A	Participant no.	Condition B	Participant no.	Condition C
1	3	5	2	9	10
2	5	6	7	10	8
3	6	7	9	11	7
4	3	8	8	12	11

14.	CO 5			10
	Apply			
17,15,14,26,9,4,19,12,9,15				
Explain the various tools for statistical quality control. Solve: An inspection of 10 samples of size 400 each from 10 lots revealed the following number of defective units.				
OR				
15.	CO 5			10
	Apply			
4.2.2				
10				
Discuss the importance of control charts. Solve : In a glass factory, the task of quality control was done with the help of mean and SD charts. 18 samples of 10 items each were chosen and then values of Σx and ΣS were found to be 595.8 and 8.28 respectively. Determine the 3 σ limits for SD chart. Given that A1=1.03, B3=0.28 and B4=1.72				

2023

SN

22IS610

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

VI Semester: TEST-1 (Section: F)

22IS610- Statistical Methods in Information Processing

Duration : 1 Hr. 15 min

Note:

- I. Part A is Compulsory.
- II. Attend any one from Part B.

Max. Marks:30

Q.N O	CO	CD	PI	QUESTION	MARK S
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PART A

A retail company has been tracking the annual sales (in thousands of dollars) across its branches. For internal reporting and strategy development, the management wants to better understand the distribution of sales figures. The data is presented showing the number of branches with annual sales below certain thresholds.

Annual Sales	Frequency
Less than 10	4
Less than 20	20
Less than 30	35
Less than 40	55
Less than 50	62
Less than 60	67

10 M

1 CO1 Apply

1.1.
1

The management has asked the data analytics team to summarize the data using appropriate central tendency measures. Calculate the **mean**, **median** and **mode** of the annual sales

2 CO2 Apply 4.2.
2

The department of Statistics at high school is evaluating the performance of students in 2 different papers of exam – Paper I and Paper II. The department wants to understand the relationship between scores in these 2 papers to see if performance in one paper can be used to predict performance in other.

5 M

Student	Paper I	Paper II
A	45	56

B	55	55
C	66	45
D	75	65
E	85	62
F	100	71

- I) Predict the paper I score for a student who scored 60 marks in paper II
 II) Write down both regression equation

PART B

Calculate the karl pearson co-efficient of skewness for the following data.

Size	1	2	3	4	5	6	7
Frequency	10	18	30	25	12	3	2

5 M

The ranks of 8 students of BE in CS and Civil on the basis of ranks is given below. To what extent the knowledge of the student in CS and Civil is related by using rank correlation co-efficient and Interpret the results.

Rank in CS	1	2	3	4	5	6	7	8
Rank in Civil	2	4	1	5	3	8	7	6

10 M

OR

Explain the concept of skewness and kurtosis with example

5 M

From the data given below, find

- a) 2 Regression co-efficient
 b) The 2 regression equation
 c) Co-efficient of correlation between marks in economics and statistics
 d) Most likely marks in statistics when marks in economics is 30.

10 M

Economic s	2	2	3	3	3	3	3	2	3	3	3	3
Statistics	5	8	5	2	1	6	9	8	4	2	3	9